5. cvičení

Příklady

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Symbol $[\cdot]$ značí celou část.

1. $\lim\limits_{n\to\infty} \dfrac{5n^3-2n+1}{-10n^3+n^2}$

2. $\lim\limits_{n\to\infty} \dfrac{(n+5)^{60}-(n+2)^{60}}{(n+3)^{60}-(n+1)^{60}}$

3. $\lim\limits_{n\to\infty} \dfrac{(n+6)^9-(n+1)^9}{(n^2+4)^5-(n^2+1)^5}$

4. $\lim\limits_{n\to\infty} \dfrac{(n+1)^{20}-n^{20}-20n^{19}}{n^{18}+n+1}$

5. $\lim\limits_{n\to\infty} \dfrac{(n^2+1)^{10}-(n+1)^{20}+20n^{19}}{1+2+3+\ldots+n^9}$

6. $\lim\limits_{n\to\infty} \left(\sqrt[3]{n^3+2n^2}-n\right)$

7. $\lim\limits_{n\to\infty} \left(\sqrt{n^2+4n+1}-\sqrt[3]{n^3+3n^2}\right)$

8. $\lim\limits_{n\to\infty} \left(\sqrt{n+\sqrt{n+\sqrt{n}}}-\sqrt{n}\right)$

9. $\lim\limits_{n\to\infty} \dfrac{\sqrt[4]{n+5}-\sqrt[4]{n+1}}{\sqrt[4]{n+9}-\sqrt[4]{n+3}}$

10. $\lim\limits_{n\to\infty} n^3\left(\sqrt[4]{n^4+1}-n\right)$

11. $\lim\limits_{n\to\infty} n^2\left(\sqrt[3]{n^3+1}-\sqrt[3]{n^3-1}\right)$

12. $\lim\limits_{n\to\infty} \dfrac{[n\sqrt{3}]}{n+[\sqrt{n}]}$

13. $\lim\limits_{n\to\infty} \sqrt[n]{n^2\cdot 2^n+n\cdot 3^n+\dfrac{4^n}{n}}$

14. $\lim\limits_{n\to\infty} \dfrac{\sqrt[n]{1^n+2^n+3^n+\ldots+n^n}}{n}$

15. $\lim\limits_{n\to\infty} \dfrac{\sqrt[3]{4^n+2^n}-\sqrt[3]{4^n-2^n}}{\sqrt[3]{4^{n+1}+2^{n+1}}-\sqrt[3]{4^{n+1}-2^{n+1}}}$

16. $\lim\limits_{n\to\infty} \dfrac{\left(1+\dfrac{1}{n}\right)^{21}-\left(1+\dfrac{30}{n^2}\right)^7-\dfrac{21}{n}}{\sqrt{n^4+20}-\sqrt{n^4+1}}\cdot\sqrt[n]{n^n+2}$

17. $\lim\limits_{n\to\infty} \dfrac{2^n\cdot\sqrt{n^2+n}-n\cdot\sqrt{4^n+1}}{\sqrt[n]{2^{n^2}+1}}$

Výsledky

1. $-\dfrac{1}{2}$ 2. $\dfrac{3}{2}$ 3. 3 4. 190 5. -360 6. $\dfrac{2}{3}$ 7. 1 8. $\dfrac{1}{2}$ 9. $\dfrac{2}{3}$ 10. $\dfrac{1}{4}$ 11. $\dfrac{2}{3}$ 12. $\sqrt{3}$ 13. 4 14. 1 15. $\sqrt[3]{2}$ 16. 140 17. $\dfrac{1}{2}$